

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA07 **COURSE CODE:** Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks.
(BL2, BL3)

Assessment Tool Weightage: 40%

SHORT ANSWER TEST

Dr.V.Parthipan

Code of Conduct:

I Antony Magrin J 192572103 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Antony Magrin J

Annexure A – Short Answer Test

1. A company needs to transfer a 150 MB design file from a client computer to a remote server over a network with a bandwidth of 50 Mbps. The Transport Layer divides the file into 1500-byte segments, while the Data Link Layer encapsulates each segment into frames by adding 40 bytes of header and trailer. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.
2. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.
3. A 12,000-byte IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is 1500 bytes. Assuming each fragment requires a 20-byte IP header, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.
4. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.
5. A multimedia application transfers a 600 MB video file. The Session Layer inserts a synchronization checkpoint after every 60 MB of transmitted data. During transmission, a failure occurs after 390 MB has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.
6. Before transmission, the Presentation Layer compresses a 900 MB file by 40%. The compressed data is encrypted, adding 5% overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.
7. An FTP application transfers 3 GB of data over a network with an effective throughput of 120 Mbps. The Application Layer divides the data into requests of 3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.
8. A packet experiences the following processing delays at different OSI layers: Application Layer = 4 ms, Transport Layer = 3 ms, Network Layer = 5 ms, Data Link Layer = 2 ms, and Physical Layer = 1 ms. The propagation delay is 18 ms, and the transmission delay is 12 ms. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9. A network transfers 80 MB of useful data using frames of 1600 bytes, where each frame contains 80 bytes of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of 240 MB through an IoT-enabled healthcare network. The Presentation Layer compresses the data by 30%, the Transport Layer divides the compressed data into 1400-byte segments, and the Data Link Layer adds 60 bytes of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear exp
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or in
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers v
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, diff understand

CSA07 – Computer Networks: Assessment Tool 1 (Short Answer Test) – Answers

Question 1

Given:

- File size = 150 MB = 150,000,000 bytes
- Bandwidth = 50 Mbps
- Segment size (Transport Layer) = 1500 bytes
- DLL overhead per frame = 40 bytes (header + trailer)

Calculations:

Number of segments = $150,000,000 / 1500 = 100,000$ segments

Since each segment becomes one frame: Number of frames = 100,000 frames

Total DLL overhead = $100,000 \times 40$ bytes = 4,000,000 bytes = 4 MB

Total frame size = $1500 + 40 = 1540$ bytes; Total data transmitted = $100,000 \times 1540 = 154,000,000$ bytes = 154 MB

Total bits transmitted = $154,000,000 \times 8 = 1,232,000,000$ bits

Transmission time = $1,232,000,000 / 50,000,000 = 24.64$ seconds

Explanation:

The Transport Layer ensures reliable communication by breaking the file into manageable segments, numbering them sequentially, and using acknowledgements, checksums, and retransmission (as in TCP) to guarantee that every segment reaches the destination in order and without errors. It also performs flow control and congestion control to avoid overwhelming the network or receiver.

The Data Link Layer ensures reliability at the link level by adding a header (source/destination MAC address, frame control) and trailer (CRC/FCS for error detection) to each frame. If a frame arrives with errors, it is discarded and can be retransmitted, ensuring error-free transfer across each physical link.

Question 2

Given:

- Sliding window size = 6 frames
- Each frame = 1024 bytes
- Total frames sent = 48

Calculations:

Number of transmission windows = Total frames / Window size = $48 / 6 = 8$ windows

If 1 frame is lost per window, retransmissions needed = 1×8 windows = 8 retransmissions

Explanation:

Flow control regulates how much data a sender can transmit before waiting for acknowledgement, using a sliding window instead of a stop-and-wait approach. This allows multiple frames to be 'in flight' at once, improving link utilization and throughput. It also prevents a fast sender from overwhelming a slow receiver's buffer, and by isolating losses to a single window, only the affected frames need retransmission rather than the entire transfer, which improves overall communication efficiency.

Question 3

Given:

- IP packet size = 12,000 bytes
- MTU = 1500 bytes
- IP header per fragment = 20 bytes

Calculations:

Usable data per fragment = $1500 - 20 = 1480$ bytes

Number of fragments = $\text{ceil}(12,000 / 1480) = \text{ceil}(8.11) = 9$ fragments

Total header overhead = $9 \times 20 = 180$ bytes

Explanation:

The Network Layer ensures successful delivery of fragmented packets using the Identification field (same value for all fragments of one packet), the Fragment Offset field (marks each fragment's position in the original packet), and the More Fragments (MF) flag (set on all fragments except the last). The destination host collects all fragments with the same identification value, reorders them using the offsets, and reassembles the original packet once the fragment with MF = 0 arrives. If any fragment is missing, the whole packet is discarded and retransmission is requested by upper layers.

Question 4 (same as Question 2)

Given:

- Sliding window size = 6 frames
- Each frame = 1024 bytes
- Total frames sent = 48

Calculations:

Number of transmission windows = $\text{Total frames} / \text{Window size} = 48 / 6 = 8$ windows

If 1 frame is lost per window, retransmissions needed = $1 \times 8 \text{ windows} = 8$ retransmissions

Explanation:

Flow control regulates how much data a sender can transmit before waiting for acknowledgement, using a sliding window instead of a stop-and-wait approach. This allows multiple frames to be 'in flight' at once, improving link utilization and throughput. It also prevents a fast sender from overwhelming a slow receiver's buffer, and by isolating losses to a

single window, only the affected frames need retransmission rather than the entire transfer, which improves overall communication efficiency.

Question 5

Given:

- Video file = 600 MB
- Checkpoint every 60 MB
- Failure occurs after 390 MB transferred

Calculations:

Checkpoints created before failure = $\text{floor}(390 / 60) = 6$ checkpoints (at 60, 120, 180, 240, 300 and 360 MB)

Data to be retransmitted after recovery = $390 - 360 = 30$ MB (data sent after the last checkpoint)

Explanation:

The Session Layer manages dialog control and synchronization between communicating systems. By inserting checkpoints, it allows a failed transfer to resume from the last successful checkpoint instead of restarting from zero. This saves time, bandwidth, and processing resources, and is especially valuable for large file or multimedia transfers where a full restart after failure would be highly inefficient.

Question 6

Given:

- Original file = 900 MB
- Compression reduces size by 40%
- Encryption adds 5% overhead to the compressed size

Calculations:

Compressed size = $900 \times (1 - 0.40) = 900 \times 0.60 = 540$ MB

Encrypted size = $540 \times 1.05 = 567$ MB

Total reduction compared to original = $(900 - 567) / 900 \times 100 = 37\%$

Explanation:

The Presentation Layer improves transmission efficiency by compressing data, which reduces the volume of data sent over the network and shortens transmission time. It improves security by encrypting the data, protecting confidentiality even though encryption itself adds a small overhead. It also handles data translation and formatting (e.g., character encoding, syntax conversion) so that data from different systems can be understood correctly at the receiving end.

Question 7

Given:

- Data to transfer = 3 GB = 3000 MB
- Effective throughput = 120 Mbps
- Each Application Layer request = 3 MB

Calculations:

Number of requests = $3000 \text{ MB} / 3 \text{ MB} = 1000$ requests

Total data in bits = $3000 \text{ MB} \times 8 = 24,000$ Megabits

Transmission time = $24,000 / 120 = 200$ seconds

Explanation:

The Application Layer provides reliable services to users by offering protocols such as FTP, HTTP, and SMTP that rely on the underlying Transport Layer (e.g., TCP) for guaranteed, in-order delivery. It manages user-level sessions and requests, structures data transfer into manageable chunks, provides authentication, and presents a simple, standardized interface so that end users and applications do not need to worry about the complexities of the underlying network.

Question 8

Given:

- Application = 4 ms, Transport = 3 ms, Network = 5 ms, Data Link = 2 ms, Physical = 1 ms
- Propagation delay = 18 ms
- Transmission delay = 12 ms

Calculations:

Total processing delay = $4 + 3 + 5 + 2 + 1 = 15$ ms

Total end-to-end delay = Processing delay + Propagation delay + Transmission delay = $15 + 18 + 12 = 45$ ms

Explanation:

Each OSI layer adds its own processing delay as the packet moves down the sender's stack and up the receiver's stack: the Application Layer prepares the data/request, the Transport Layer handles segmentation and reliability checks, the Network Layer performs routing and addressing decisions, the Data Link Layer handles framing and error checking, and the Physical Layer converts data to signals. In addition, transmission delay (time to push all bits onto the link) and propagation delay (time for a signal to travel across the medium) contribute to the total end-to-end delay experienced by the packet.

Question 9

Given:

- Useful data = 80 MB = 80,000,000 bytes
- Frame size = 1600 bytes
- Overhead per frame = 80 bytes

Calculations:

Useful data per frame = $1600 - 80 = 1520$ bytes

Number of frames = $\text{ceil}(80,000,000 / 1520) = 52,632$ frames

Total overhead = $52,632 \times 80 = 4,210,560$ bytes ≈ 4.21 MB

Total data transmitted (with overhead) = $52,632 \times 1600 = 84,211,200$ bytes

Transmission efficiency = $(\text{Useful data} / \text{Total data transmitted}) \times 100 = (80,000,000 / 84,211,200) \times 100 \approx 95.0\%$

Explanation:

Protocol overhead is the extra header/trailer data added by different OSI layers for addressing, error checking, and control purposes. While necessary for reliable communication, overhead reduces the proportion of a frame that carries actual useful data, lowering effective throughput. Networks aim to balance frame size and overhead — very small frames waste more bandwidth on overhead, while very large frames increase the impact of errors, since a single bit error forces retransmission of the whole frame.

Question 10**Given:**

- Medical record = 240 MB
- Presentation Layer compression = 30%
- Transport Layer segment size = 1400 bytes
- Data Link Layer overhead = 60 bytes per frame

Calculations:

Compressed size = $240 \times (1 - 0.30) = 240 \times 0.70 = 168$ MB = 168,000,000 bytes

Number of segments = $168,000,000 / 1400 = 120,000$ segments

Total Data Link overhead = $120,000 \times 60 = 7,200,000$ bytes = 7.2 MB

Final data transmitted across the Physical Layer = Compressed data + DLL overhead = $168,000,000 + 7,200,000 = 175,200,000$ bytes ≈ 175.2 MB

Explanation:

In this IoT-enabled healthcare scenario, multiple OSI layers work together to ensure secure and reliable communication. The Presentation Layer compresses (and can encrypt) the sensitive medical data to reduce size and protect confidentiality. The Transport Layer divides the compressed data into fixed-size segments and is responsible for reliable, ordered, end-to-end delivery. The Network Layer routes these segments across the IoT network to the correct destination. The Data Link Layer frames each segment, adding overhead for addressing and error detection so that transmission errors on any single link can be identified and corrected. Together, these layers ensure the patient's medical record reaches its destination completely, accurately, and securely.